Europes first web page and database for all shelter dogs & cats in Europe

This could radically increase adoption rates and animal welfare in Europe for shelter dogs and cats and avoid long-term suffering in the shelters or being euthanized. Most people only look at the shelter webpages in their own country when they are looking to adopt a cat or dog and if they don't find a perfect match then they buy a dog or cat instead of adopting.

Transport and documents within Europe are very cheap and simple, it is really easy to move animals from one EU country to another by car, plane or ship. Also, there is an imbalance between western / northern Europe and eastern / south Europe. Eastern and southern Europe have many animals in their shelters but a lack of potential adopters while the western and northern Europe have few animals in shelters but a lot of potential adopters.

Having a central European webpage for all shelter animals in Europe would result in people being able to look at thousands of different dogs and cats and most likely find that perfect match for them - their new best friend, family member, fur-baby and in some cases, their soulmate.

For European dogs and cats to find their ideal home and for European homes to find their ideal pet.

Local shelters do not need to worry that this will result in fewer adoptions nationwide since people will always prefer to adopt an animal from a shelter close to them that they can meet before adoption. This option is meant to be the backup if people don't find an animal in a shelter close to them.

The front-end of the webpage

The “start” page will present a simple and easy way of searching for a dog or cat. The search criterias for dogs will be: sex, age, size, in all EU or a specific country and for cats: sex, age, all EU or a specific country.

Something similar to: https://varjupaik.ee/loomad/loomad-varjupaigas.

Once a user clicks on a dog or cat they will also see “more info” where the shelter writes any additional info about the animal (personality, history, special needs etc) and medical info (sterilized/castrated, vaccines). The user will also see which country and shelter the

animal is located in (contact info) and a link to an interest form they can fill in that will automatically be sent to the shelter.

This structure also makes it possible for people to see all dogs and cats in their country. National overviews of all shelter animals do not exist in most countries. So our webpage would cover the lack of an European webpage / database and also the lack of national webpages / databases.

On the upper menu that is located in the top part of the webpage horizontally there will be the choices: Start, Our mission, Donate, Useful info & links (transportation, documents, why shelter animals are a safe option etc) and Happy stories (success stories of adoptions).

At the top of the webpage there will also be language options. All European languages should be represented here. Language barriers should not stop people from adopting from another country. It would be ideal if we could offer a chat option on our webpage for the shelter and the adopter, similar to airbnb-s chat option where users can also choose the language they want to communicate in and then everything will be automatically translated into that language. Many European people are still not comfortable communicating in english.

Back-end and database

This project needs to use a relational database (which one is most fitted for this project?). We need to take into account that this database will most likely eventually store the information about thousands of dogs and cats. Chat gpt recommends to use Postgre SQL - “We need to consider scalability, performance, and support for complex queries. A PostgreSQL database is an excellent choice for this project because it is robust, supports advanced search features, and handles large datasets efficiently.”

An example of the database design and architecture:

1. Animals

Stores information about individual animals.

Column	Data Type	Description
`animal_id`	SERIAL PRIMARY KEY	Unique identifier for the animal.
`name`	VARCHAR(100)	Name of the animal.

species	ENUM ('dog', 'cat')	Whether the animal is a dog or cat.
sex	ENUM ('male', 'female')	Gender of the animal.
age	SMALLINT	Age in years.
size	ENUM ('small', 'medium', 'large')	Size of the animal (only for dogs).
description	TEXT	Additional info about the animal (e.g., personality, needs).
medical_info	TEXT	Medical info (e.g., sterilized, vaccinations).
status	ENUM ('available', 'adopted', 'in treatment')	Current status of the animal.
shelter_id	INT	Foreign Key linking to the shelter where the animal is housed.
created_at	TIMESTAMP	Record creation timestamp.

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2. Shelters

Stores information about shelters.

Column	Data Type	Description
shelter_id	SERIAL PRIMARY KEY	Unique identifier for the shelter.
name	VARCHAR(150)	Name of the shelter.
location	VARCHAR(255)	Full address of the shelter.
country	VARCHAR(100)	Country where the shelter is located.
contact_number	VARCHAR(20)	Shelter contact number.
email	VARCHAR(100)	Shelter email address.
interest_form_url	TEXT	URL of the interest form for adoption inquiries.

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3. Users

Tracks visitors who fill out interest forms.

Column	Data Type	Description
`user_id`	SERIAL PRIMARY KEY	Unique identifier for the user.
`name`	VARCHAR(100)	Name of the user.
`email`	VARCHAR(100)	User's email address.
`phone`	VARCHAR(20)	Contact number.
`message`	TEXT	Message or inquiry from the user.
`animal_id`	INT	Foreign Key linking to the animal the inquiry is about.

4. Happy Stories

Stores success stories about adoptions.

Column	Data Type	Description
`story_id`	SERIAL PRIMARY KEY	Unique identifier for the story.
`title`	VARCHAR(150)	Title of the story.
`content`	TEXT	The story content.
`image_url`	TEXT	URL of an image related to the story.
`created_at`	TIMESTAMP	Timestamp of story creation.

Most likely this example needs to be developed more. The interest forms that the adopters fill out needs to be automatically sent out to the shelters : Interest Form Submission: when a user submits an interest form, it will be saved to the `users` table and optionally trigger an email to the shelter.

Scalability Considerations:

1. Use indexes on frequently queried columns (species, sex, size, age).
2. Partition the animals table by species or country to handle large datasets efficiently.
3. Regularly archive old data (e.g., adopted animals).

The goal is to have the webpage automated (automation) as much as possible to minimize the need for maintenance and administration. The goal is to only have tech support if the shelters have any technical issues.

Key Steps for Implementation in Real Systems:

1. Automated Processes:
    - Scheduled scripts for syncing data (e.g., cron jobs or cloud functions).
    - Automatic updates to the database from shelter-provided APIs or forms.
2. Error Handling:
    - Use centralized logging systems like ELK stack (Elasticsearch, Logstash, Kibana) or CloudWatch.
    - Alert administrators only when critical issues occur.
3. Self-Service for Shelters:
    - Build a dashboard with role-based access for shelters to manage their own data.
4. Minimal Tech Support:
    - Ensure the platform is well-documented with a robust FAQ section for shelter staff.
    - Automate responses for common technical queries using chatbots.

This approach combines automation, robust error handling, and scalability to ensure the system operates with minimal manual intervention.

Webpage names:

The following domain names are available for registering:

www.petsofeurope.org or .eu
www.dogsofeurope.org or.eu
www.catsofeurope.org or .eu

After we are ready with the webpage:

I also have plans for what needs to be done before launching the webpage, the financial part, getting sponsors, EU grants and donations, advertising, getting all European shelters to sign up and upload all dogs and cats and much more.

But one step at a time, let's make the webpage first ☺ !